



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



YEAR 12 2008 HALF YEARLY EXAMINATIONS

MATHEMATICS EXTENSION 1
(3 UNIT COMPONENT)

ASSESSMENT TASK 3
WEIGHTING 30%

Examination Date:
Friday 2nd May 2008

OUTCOMES REFERRED TO: HE1, HE2, HE3, HE4, HE6, HE7, PE1, PE2, PE3, PE4, PE6.

General Instructions

- Reading time – **5 minutes**.
- Working time – **2 hours**.
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- All necessary working should be shown in every question.
- **Begin** each question in a **new booklet**.
- Write your **examination number** and your **teacher's name** on the front of each answer booklet.

Total marks – 84

- Attempt Questions 1-7.
- All questions are of equal value.
- Mark values are shown at the side of each question part.

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QUESTION 1 (12 marks) Use a SEPARATE writing booklet.**Marks**

- (a) Solve the inequation $\frac{4}{5-x} \leq 1$. 3
- (b) Use the substitution $u = 1 + \tan x$ to evaluate $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{\sqrt{1 + \tan x}} dx$. 3
- (c) Find the following:
- i) $\int \frac{3x}{x^2 + 9} dx$. 2
- ii) $\int \frac{3}{x^2 + 9} dx$. 2
- iii) $\int \frac{x^2 + 9}{3x} dx$. 2

QUESTION 2 (12 marks) Use a SEPARATE writing booklet.**Marks**

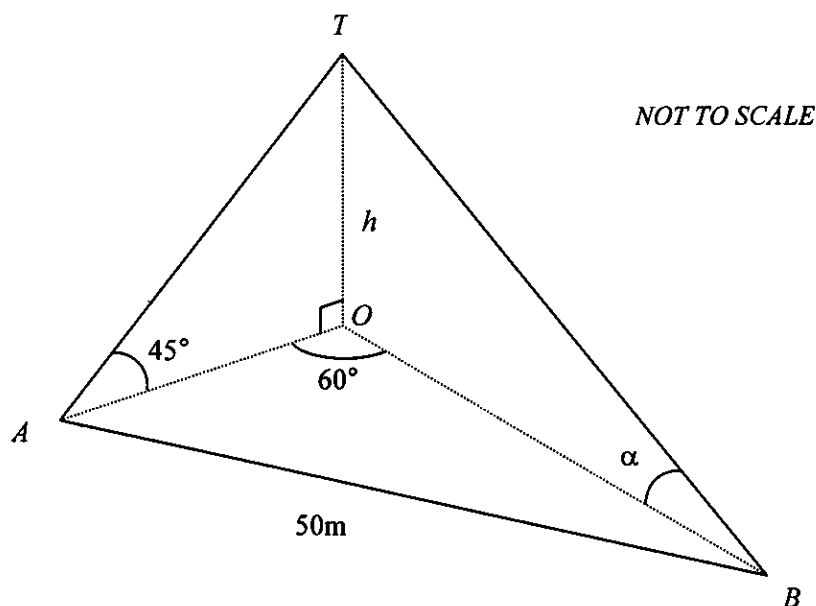
- (a) Solve $\frac{x+1}{x^2+1} > 1$. 2
- (b) Find $\int x(1-x^2)^5 dx$, using the substitution $u = 1-x^2$, or otherwise. 2
- (c) Find the acute angle, correct to the nearest minute, between the lines $3x + y = 4$ and $x - y = 1$. 2
- (d) The point $P(19, -15)$ divides an interval AB externally in the ratio $3:2$. Find the coordinates of the point $B(x, y)$ given $A(-2, 3)$. 3
- (e) Prove by mathematical induction that if n is a positive integer, then: 3
- $$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}.$$

QUESTION 3 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) i) If $f(x) = e^{x+2}$, find the inverse function $f^{-1}(x)$. 2
- ii) On the same axes, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$. 2
- (b) Differentiate $x^2 \cos^{-1}(3x)$. 2
- (c) Find the exact value of $\cos^{-1}\left(\sin \frac{4\pi}{3}\right)$. 3
- (d) Determine the domain and range of $y = 2 \sin^{-1}\left(\frac{x}{3}\right)$ and hence sketch the graph. 3

(a)



In the diagram, the points A , B and O are in the same horizontal plane.

A and B are 50m apart and $\angle AOB = 60^\circ$. OT is a vertical tower of height h metres.

The angles of elevation of T from A and B respectively are 45° and α .

(α is acute.)

- i) Show $AO = h$. 1
 - ii) Show $OB = h \cot \alpha$. 1
 - iii) By using the cosine rule in triangle AOB , show that: 1

$$h^2 \cot^2 \alpha - h^2 \cot \alpha + h^2 = 50^2.$$
 - iv) Given the tower is 30 high, find the angle α correct to the nearest degree. 2
- (b) Write down the general solution, in terms of π , of the equation $\tan \theta = -\frac{1}{\sqrt{3}}$. 2
- (c) α and β are acute angles such that $\cos \alpha = \frac{3}{5}$ and $\sin \beta = \frac{1}{\sqrt{5}}$. 2
 Without finding the size of either angle, show that $\alpha = 2\beta$.
- (d) Without using a calculator, find the exact value of $\sin\left(\cos^{-1}\frac{2}{3} + \sin^{-1}\frac{1}{4}\right)$. 3

QUESTION 5 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Newton's Law of Cooling states that when an object at temperature T ($^{\circ}\text{C}$) is placed in an environment at a temperature R ($^{\circ}\text{C}$), then the rate of temperature loss is given by the equation $\frac{dT}{dt} = k(T - R)$; where t is the time in seconds and k is a constant.

A packet of peas, initially at 24°C is placed in a snap-freeze refrigerator in which the internal temperature is maintained at -40°C . After 5 seconds the temperature of the packet is 19°C . Suppose $T = R + Ae^{kt}$, where A is a constant.

- i) State the value of A . 2
- ii) Show that $k = \frac{1}{5} \log_e \left(\frac{59}{64} \right)$. 2
- iii) Hence show that it will take approximately 29 seconds for the packet's temperature to reduce to 0°C . 2

Question 5 continues on page 7.

QUESTION 5 (continued)

(b) If $y = \frac{1}{200} t e^{-t}$, show that $\frac{dy}{dt} = \frac{1}{200} (1 - t) e^{-t}$. **1**

- (c) Kiran has recently consumed three standard alcoholic drinks.
Immediately after he has finished his last drink, his blood alcohol level
is measured over a four-hour period.

Let his blood alcohol level at any time t be A , where t is the time in hours
after his last drink.

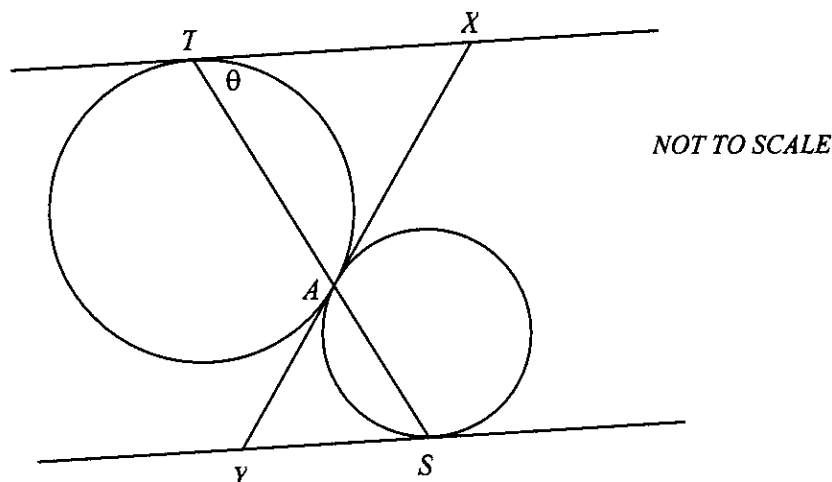
It is found that the rate of change $\frac{dA}{dt}$ of his blood alcohol content is given by:

$$\frac{dA}{dt} = \frac{1}{200} (1 - t) e^{-t}, \text{ where } 0 \leq t \leq 4.$$

- i) Show that his blood alcohol content increases during the first hour **2**
and decreases after the first hour.
- ii) Initially his blood alcohol content was 0.0005 . Find A as a function of t . **2**
You will need to use part (b).
- iii) Determine his maximum alcohol content during the four-hour period. **1**
Give your answer correct to four decimal places.

End of Question 5.

(a)



In the diagram above, two circles touch one another externally at the point A .

A straight line through A meets one of the circles at T and the other at S .

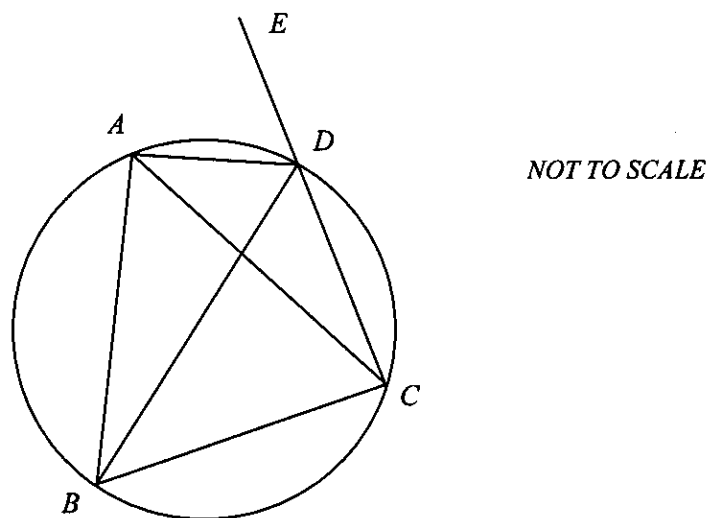
The tangents at T and S meet the common tangent at A at X and Y respectively.

Let $\theta = \angle XTA$.

i) Explain why $\angle XAT$ is θ . 1

ii) Prove that $TX \parallel YS$. 2

(b)



$ABCD$ is a cyclic quadrilateral in which $AB = AC$, and CD is produced to E .

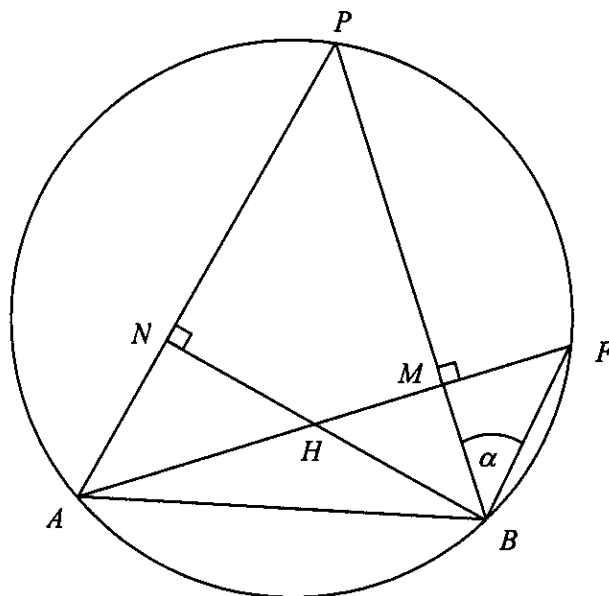
Prove that AD bisects $\angle BDE$. (Hint: let $\angle ADE = \theta$).

3

Question 6 continues on page 9.

QUESTION 6 (continued)

(c)



NOT TO SCALE

In the diagram above, ABP is a triangle inscribed in a circle.

The altitudes BN and AM of the triangle intersect at H .

The altitude AM is produced to meet the circumference of the circle at F .

Copy the diagram into your examination booklet.

Let $\angle PBF = \alpha$.

- | | | |
|------|---|---|
| i) | Why is $\angle PAF = \alpha$? | 1 |
| ii) | Why are points A , N , M and B concyclic? | 1 |
| iii) | Why is $\angle NBM = \alpha$? | 1 |
| iv) | Show that M bisects HF . | 3 |

End of Question 6.

QUESTION 7 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$ ($a > 0$).

i) By derivation, show that the equation of the chord is: **2**

$$y = \frac{1}{2}(p+q)x - apq.$$

ii) If the chord PQ passes through the focus, S , show that $pq = -1$. **1**

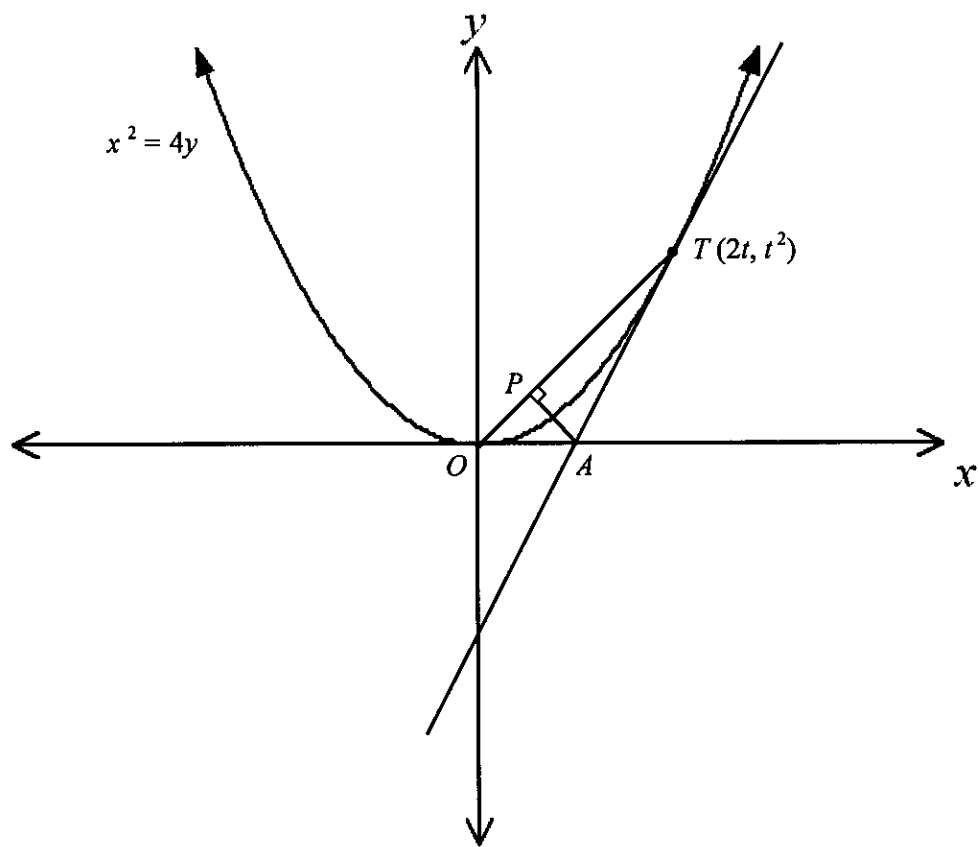
iii) Using the fact that $PQ = PS + SQ$, or otherwise, show that **3**

the chord PQ has length $a\left(p + \frac{1}{p}\right)^2$.

Question 7 continues on page 11.

QUESTION 7 (continued)

(b)



The tangent at $T(2t, t^2)$, $t \neq 0$, on the parabola $x^2 = 4y$ meets the x -axis at A .

$P(x, y)$ is the foot of the perpendicular from A to OT , where O is the origin.

The equation of the tangent at T is $y = tx - t^2$.

- i) Prove that the equation of AP is $y = -\frac{2}{t}(x - t)$. 2
- ii) Show that the equation of OT is $t = \frac{2y}{x}$. 1
- iii) Hence, or otherwise, prove that the locus of $P(x, y)$ lies on a circle 3
with centre $(0, 1)$ and give its radius.

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SOLUTIONS

* MATHEMATICS EXTENSION 1 (3 Unit component) YEAR 12 - 2008 - HALF YEARLY EXAMINATION

Question 1

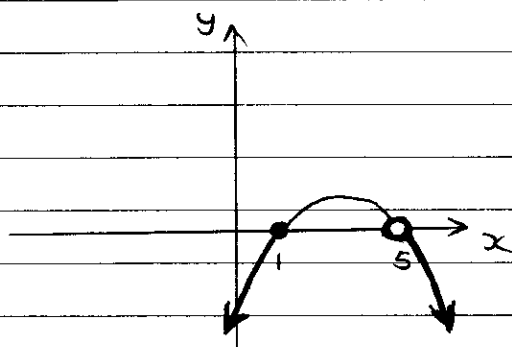
a) $\frac{4}{5-x} \leq 1 \rightarrow x \neq 5$ (can't ÷ by 0)

$$(5-x)^2 \frac{4}{5-x} \leq (5-x)^2 \quad (1)$$

$$(5-x)[4 - (5-x)] \leq 0$$

$$(5-x)(x-1) \leq 0 \quad (1)$$

$$\therefore x \leq 1 \text{ or } x > 5 \quad (1)$$



b) $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{\sqrt{1+\tan x}} dx$

$$= \int_1^2 \frac{1}{\sqrt{u}} du \quad (1)$$

$$= \left[2u^{\frac{1}{2}} \right]_1^2 \quad (1)$$

$$= 2\sqrt{2} - 2 \quad (1)$$

Let $u = 1 + \tan x$

$$\therefore \frac{du}{dx} = \sec^2 x$$

when $x = 0 \rightarrow u = 1$

$x = \frac{\pi}{4} \rightarrow u = 2$

c) i) $\int \frac{3x}{x^2+9} dx = \frac{3}{2} \int \frac{2x}{x^2+9} dx \quad (1)$

$$= \frac{3}{2} \ln(x^2+9) + C \quad (1)$$

$$c) ii) \int \frac{3}{x^2+9} dx = 3 \int \frac{1}{9+x^2} dx$$

$$= 3 \times \frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) + C \quad (1)$$

$$= \tan^{-1}\left(\frac{x}{3}\right) + C \quad (1)$$

[full marks
for straight
to answer
in one step
also]

$$iii) \int \frac{x^2+9}{3x} dx = \int \frac{x^2}{3x} + \frac{9}{3x} dx$$

$$= \int \frac{x}{3} + \frac{3}{x} dx \quad (x \neq 0) \quad (1)$$

$$= \frac{x^2}{6} + 3 \ln x + C \quad (1)$$

Question 2

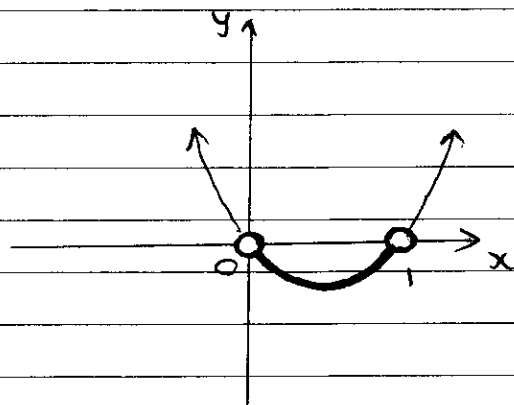
$$a) \frac{x+1}{x^2+1} > 1$$

$$x+1 > x^2+1 \quad (1) \quad (\text{as } x^2+1 > 0)$$

$$x^2 - x < 0$$

$$x(x-1) < 0$$

$$\therefore 0 < x < 1 \quad (1)$$



$$b) \int x(1-x^2)^5 dx$$

$$= -\frac{1}{2} \int -2x(1-x^2)^5 dx \quad (1)$$

$$= -\frac{1}{2} \int u^5 du$$

$$= -\frac{u^6}{12} + C$$

$$= -\frac{(1-x^2)^6}{12} + C \quad (1)$$

$$\text{Let } u = 1 - x^2$$

$$\therefore \frac{du}{dx} = -2x$$

$$c) \quad \begin{aligned} 3x + y &= 4 \rightarrow m_1 = -3 \\ x - y &= 1 \rightarrow m_2 = 1 \end{aligned}$$

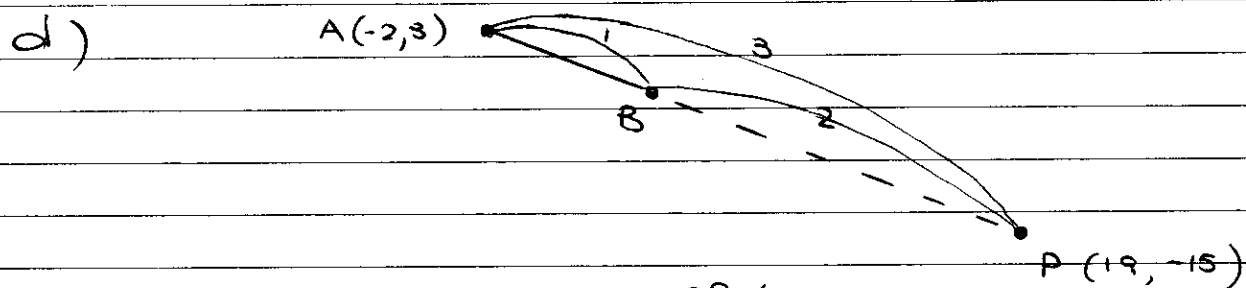
$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$= \frac{-3 - 1}{1 + (-3)(+1)} \quad (1)$$

$$= \frac{-4}{-2}$$

$$= 2$$

$$\begin{aligned} \therefore \theta &= \tan^{-1}(2) \\ &= 63^\circ 26' \text{ (to nearest min.)} \quad (1) \end{aligned}$$



METHOD 1

x-direction:

$$x_B = -2 + \frac{1}{3}(19 - (-2)) \quad (1)$$

$$= 5$$

y-direction:

$$y_B = 3 - \frac{1}{3}(3 - (-15)) \quad (1)$$

$$= -3$$

$$\therefore B(5, -3) \quad (1)$$

OR

METHOD 2

$$x = \frac{mx_2 + nx_1}{m+n}$$

$$19 = \frac{-3(x) + 2(-2)}{-3+2} \quad (1)$$

$$-19 = -3x - 4$$

$$\therefore x = 5$$

$$y = \frac{my_2 + ny_1}{m+n}$$

$$-15 = \frac{-3y + 2(3)}{-3+2} \quad (1)$$

$$15 = -3y + 6$$

$$\therefore y = -3$$

$$\therefore B(5, -3) \quad (1)$$

c) Step 1: Check first case (i.e. $n=1$)

$$\text{LHS} = \frac{1}{1 \times 5}$$

$$= \frac{1}{5}$$

$$\text{RHS} = \frac{1}{4 \times 1 + 1}$$

$$= \frac{1}{5}$$

$$= \text{LHS} \quad \therefore \text{true for } n=1$$

Step 2: Assume true for $n=k$

$$\therefore \frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4k-3)(4k+1)} = \frac{k}{4k+1}$$

Step 3: Prove true for $n=k+1$

$$\text{Q. required to prove } \frac{k}{4k+1} + \frac{1}{(4(k+1)-3)(4(k+1)+1)} = \frac{k+1}{4(k+1)+1}$$

$$\text{LHS} = \frac{k}{4k+1} + \frac{1}{(4(k+1)-3)(4(k+1)+1)} \quad (1)$$

$$= \frac{k}{4k+1} + \frac{1}{(4k+1)(4k+5)}$$

$$= \frac{k(4k+5) + 1}{(4k+1)(4k+5)}$$

$$= \frac{4k^2 + 5k + 1}{(4k+1)(4k+5)}$$

$$= \frac{(4k+1)(k+1)}{(4k+1)(4k+5)} \quad (4k+1 \neq 0)$$

$$= \frac{k+1}{4(k+1)+1} \quad \text{as req'd.} \quad (2)$$

Step 4: If true for $n=k$, then true for $n=k+1$.

Since true for $n=1$, then true for $n=2, n=3, n=4$ and so on...

$\therefore$ True for all positive integer values of n .

① all stages of induction completed correctly.

Question 3

a) i) $f(x) = e^{x+2}$

$$x = e^{y+2}$$

$$\ln x = y + 2$$

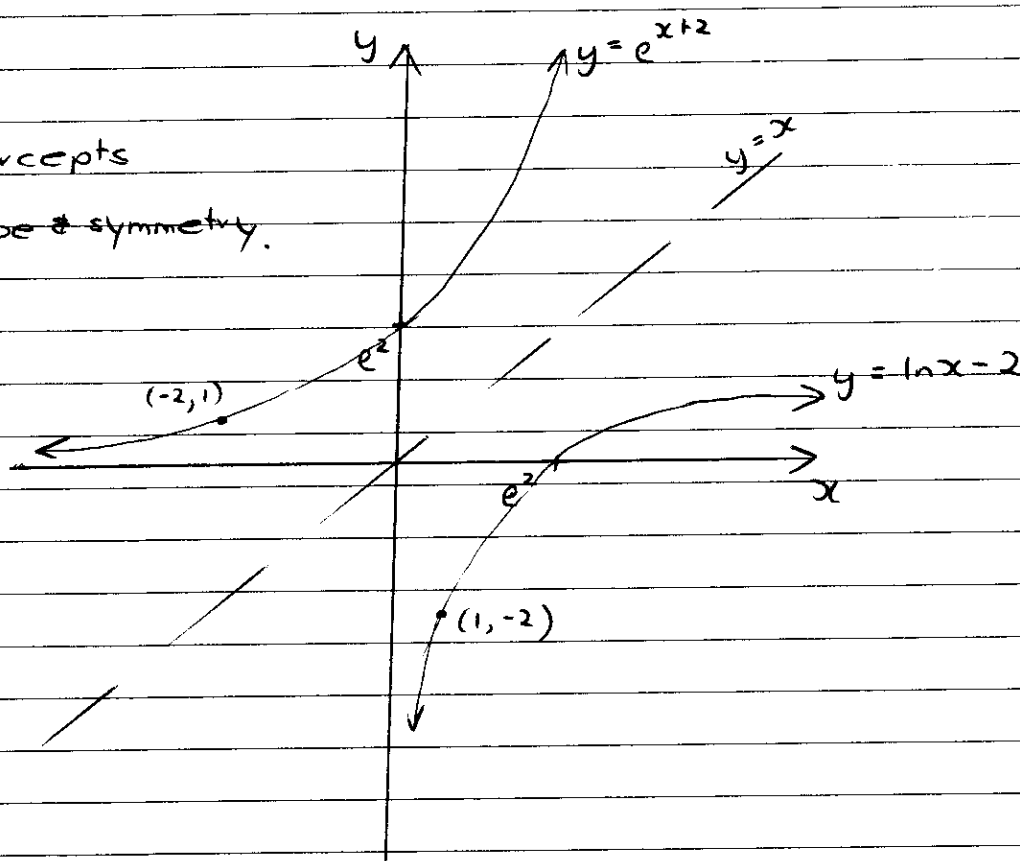
$$y = \ln x - 2$$

$$\therefore f^{-1}(x) = \ln x - 2$$

ii)

① intercepts

① shape & symmetry.



b) $\frac{d}{dx} (x^2 \cos^{-1}(3x)) = x^2 \cdot \frac{-1}{\sqrt{1-9x^2}} \cdot 3 + \cos^{-1}(3x) \cdot 2x$

$$= 2x \cos^{-1}(3x) - \frac{3x^2}{\sqrt{1-9x^2}}$$

①

①

c) Let $\alpha = \cos^{-1}\left(\sin \frac{4\pi}{3}\right)$
 (where $0 \leq \alpha \leq \pi$)

$$\therefore \cos \alpha = -\frac{\sqrt{3}}{2} \quad (1)$$

* S	A
* T	C

$\therefore \alpha$ in 2nd quadrant (1)

Related $\angle = \frac{\pi}{6}$

$$\begin{aligned} \therefore \alpha &= \pi - \frac{\pi}{6} \\ &= \frac{5\pi}{6} \quad (1) \end{aligned}$$

d) $y = 2 \sin^{-1}\left(\frac{x}{3}\right)$

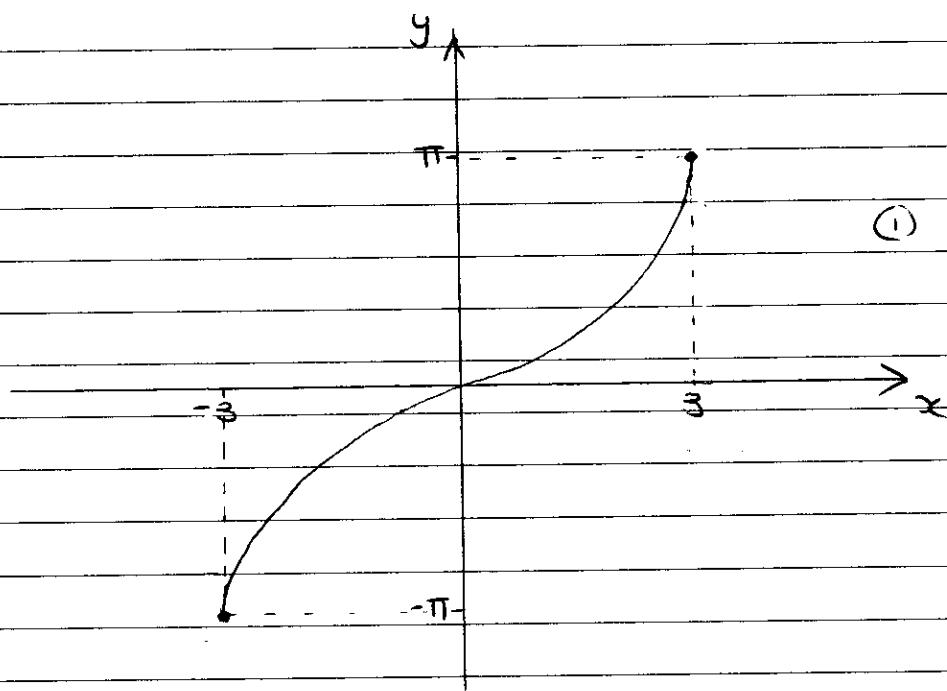
* domain: $-1 \leq \frac{x}{3} \leq 1$

$$\therefore -3 \leq x \leq 3 \quad (1)$$

* range: $-\frac{\pi}{2} \leq \sin^{-1}\left(\frac{x}{3}\right) \leq \frac{\pi}{2}$

$$-\pi \leq 2 \sin^{-1}\left(\frac{x}{3}\right) \leq \pi$$

$$\therefore -\pi \leq y \leq \pi \quad (1)$$



Question 4

a) i) $\tan 45^\circ = \frac{h}{AO}$

$$\therefore AO = \frac{h}{\tan 45^\circ}$$

$$= h$$

①

(or through
(SOS, Δ))

ii) $\tan \alpha = \frac{h}{OB}$

$$\therefore OB = \frac{h}{\tan \alpha}$$

$$= h \cot \alpha$$

①

iii) $AB^2 = AO^2 + OB^2 - 2 \cdot AO \cdot OB \cdot \cos 60^\circ$

$$50^2 = h^2 + h^2 \cot^2 \alpha - 2 \cdot h \cdot h \cot \alpha \cdot \frac{1}{2}$$

①

$$= h^2 + h^2 \cot^2 \alpha - h^2 \cot \alpha$$

Rearranging:

$$h^2 \cot^2 \alpha - h^2 \cot \alpha + h^2 = 50^2 \text{ as req'd.}$$

iv) $30^2 \cot^2 \alpha - 30^2 \cot \alpha + 30^2 = 50^2$

$$900 \cot^2 \alpha - 900 \cot \alpha + 900 = 2500$$

$$9 \cot^2 \alpha - 9 \cot \alpha + 16 = 0$$

①

$$\therefore \cot \alpha = \frac{9 \pm \sqrt{81 - 4 \times 9 \times 16}}{18}$$

$$= \frac{9 \pm \sqrt{657}}{18}$$

$$\therefore \alpha = 27^\circ \text{ (to nearest degree)}$$

①

(as $\alpha > 0$)

b) $\tan \theta = -\frac{1}{\sqrt{3}}$

S	A
T	C*

$$\text{related } \angle = \frac{\pi}{6}$$

$$\therefore \theta = n\pi - \frac{\pi}{6} \text{ where } n \text{ is an integer}$$

or

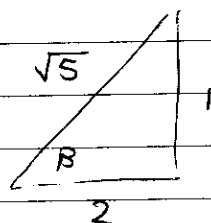
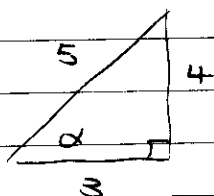
$$\theta = n\pi + \frac{5\pi}{6} \text{ where } n \text{ is an integer}$$

① expression
stylei.e. $n\pi + \dots$ or $n\pi - \dots$

① correct angle

$$c) \cos \alpha = \frac{3}{5}$$

$$\sin \beta = \frac{1}{\sqrt{5}}$$



$$\sin \alpha = \frac{4}{5} \quad (1)$$

$$\begin{aligned} \sin 2\beta &= 2 \sin \beta \cos \beta \\ &= 2 \times \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}} \end{aligned}$$

$$= \frac{4}{5} \quad (1)$$

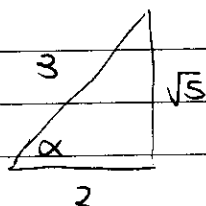
$$= \sin \alpha$$

$$\therefore \alpha = 2\beta$$

$$d) \sin \left(\cos^{-1} \frac{2}{3} + \sin^{-1} \frac{1}{4} \right)$$

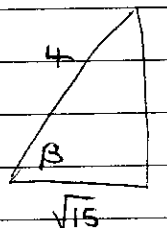
$$\text{Let } \alpha = \cos^{-1} \frac{2}{3} \quad (0 < \alpha < \pi)$$

$$\therefore \cos \alpha = \frac{2}{3} \quad \begin{array}{c} \text{S} \\ \text{---} \\ \text{T} \end{array} \begin{array}{c} \text{A}^* \\ \text{---} \\ \text{C} \end{array} \quad \therefore \text{1st quad.}$$



$$\text{Let } \beta = \sin^{-1} \frac{1}{4} \quad \left(-\frac{\pi}{2} < \beta < \frac{\pi}{2} \right)$$

$$\therefore \sin \beta = \frac{1}{4} \quad \begin{array}{c} \text{S} \\ \text{---} \\ \text{T} \end{array} \begin{array}{c} \text{A}^* \\ \text{---} \\ \text{C} \end{array} \quad \therefore \text{1st quad.} \quad (1)$$



$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta \quad (1)$$

$$= \frac{\sqrt{5}}{3} \cdot \frac{\sqrt{15}}{4} + \frac{2}{3} \cdot \frac{1}{4}$$

$$= \frac{\sqrt{75} + 2}{12}$$

$$= \frac{5\sqrt{3} + 2}{12} \quad (1)$$

Question 5

a) i) When $t=0$, $T=24^{\circ}\text{C}$, $R=-40^{\circ}\text{C}$ (1)
 $\therefore 24 = -40 + Ae^0$ (1)
 $\therefore A = 64$

ii) $t=5$, $T=19^{\circ}\text{C}$ (1)
 $\therefore 19 = -40 + 64e^{5k}$ (1)
 $\therefore e^{5k} = \frac{59}{64}$

$$\ln \frac{59}{64} = 5k$$
 (1)

$$\therefore k = \frac{1}{5} \ln \frac{59}{64}$$

iii) $t=?$ when $T=0^{\circ}\text{C}$:

$$0 = -40 + 64e^{kt}$$
 (1)

$$e^{kt} = \frac{40}{64}$$

$$\ln\left(\frac{5}{8}\right) = kt$$

$$\therefore t = \frac{\ln\left(\frac{5}{8}\right)}{\frac{1}{5} \ln\left(\frac{59}{64}\right)}$$
 (1)

$$= 28.88929\dots$$

$$\approx 29 \text{ seconds.}$$

b) $y = \frac{1}{200} te^{-t}$

$$\frac{dy}{dt} = \frac{1}{200} (t \cdot -e^{-t} + e^{-t} \cdot 1)$$

$$= \frac{1}{200} (1-t) e^{-t}$$
 (1)

c) i) when $t < 1$ $\frac{dA}{dt} > 0$ (since $(1-t) > 0$) (1)

$\therefore A$ is increasing for $t < 1$

when $t > 1$ $\frac{dA}{dt} < 0$ (since $(1-t) < 0$) (1)

$\therefore A$ is decreasing for $t > 1$

ii) When $t = 0$, $A = 0.0005$

$$A = \int \frac{dA}{dt} dt$$

$$= \int \frac{1}{200} (1-t) e^{-t} dt$$

$$= \frac{1}{200} t e^{-t} + C \quad (\text{from part b}) \quad (1)$$

when $t = 0$: $0.0005 = 0 + C$

$$\therefore C = 0.0005$$

$$\therefore A = \frac{1}{200} t e^{-t} + 0.0005 \quad (1)$$

iii) from i) the maximum occurs when $t = 1$:

$$\therefore A_{\max} = \frac{1}{200} \times 1 \times e^{-1} + 0.0005$$

$$= 0.002339397 \dots$$

$$\doteq 0.0023 \text{ (to 4 dec. pl.)} \quad (1)$$

$$\therefore TX = XA$$

So $\triangle TXA$ is isosceles and $\angle XTA = \angle XAT = \theta$
(base angles of isosceles $\triangle$)

ii) similarly, $\triangle AYS$ is isosceles with base angles $\angle YAS$ & $\angle YSA$ equal.

But $\angle YAS = \angle TAX = 90^\circ$ (vertically opposite)

So $\angle YSA = \theta$ & $\angle XTA = \theta$

But these are alternate

So TX || YS

b) Let $\angle ADE = \theta$

$\therefore \angle ABC = \theta$ (exterior angle cyclic quadrilateral)

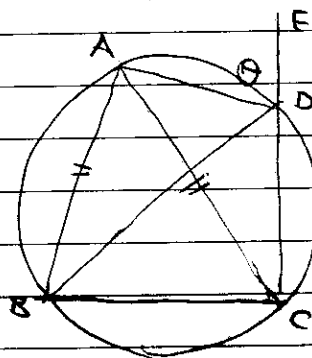
$\triangle ABC$ is isosceles ($AB = AC$, given)

$\therefore \angle ABC = \angle ACB = \theta$ (base angles isosceles Δ)

$$\angle ACB = \angle ADB = \theta \text{ (angles in same segment)}$$

$\therefore \angle ADE = \angle ADB$ (both equal θ)

$\therefore DB$ bisects $\angle BDE$



c) i) $\angle PAF = \angle PBF = \alpha$ (angles in same segment) ①

ii) $\angle ANB = \angle AMB = 90^\circ$ (given)

$\therefore AB$ is a diameter of circle through $A, N, M \neq B$

(angle in a semicircle is a right $\angle$)

$\therefore A, N, M \& B$ are concyclic.

For angles in same segment / standing on chord AB....

iii) $\angle NBM = \angle MAN$ (angles standing on same arc of circle ANMB) ①

$$\angle MAN = \angle PAF \quad (\text{same angle})$$

$$= \alpha$$

$$\therefore \angle NBM = \alpha$$

iv) In Δ 's HMB & FMB:

① { side: MB is common
angle: $\angle MBH = \angle MBF = \alpha$ (accept as proven above)
($\angle MBH$ is same as $\angle NBM$
 $\angle MBF$ is same as $\angle PBF$)

① { angle: $\angle HMB = \angle FMB = 90^\circ$ (AM is altitude of ΔAPB)
 $\therefore \Delta HMB \equiv \Delta FMB$ (AAS)

① { $\therefore HM = FM$ (matching sides of congruent Δ 's)
 $\therefore M$ bisects HF. (another option ... using $\tan \alpha = \dots$)

Question 7

$$\begin{aligned} \text{a) i) } m &= \frac{ap^2 - aq^2}{2ap - 2aq} \\ &= \frac{\cancel{a}(p+q)(\cancel{p-q})}{2\cancel{a}(\cancel{p-q})} \quad (a \neq 0 \text{ \& } p-q \neq 0) \\ &= \frac{p+q}{2} \quad \text{①} \end{aligned}$$

$$y - y_1 = m(x - x_1)$$

$$y - ap^2 = \frac{p+q}{2}(x - 2ap) \quad \text{①}$$

$$\therefore y = \frac{p+q}{2}x - \frac{2ap^2}{2} - \frac{2apq}{2} + ap^2$$

$$\therefore y = \frac{1}{2}(p+q)x - apq \quad \text{as req'd.}$$

ii) Focus $S(0, a) \rightarrow$ satisfies equation if on chord:

$$\therefore a = \frac{1}{2}(p+q) \times 0 - apq \quad \text{①}$$

$$\therefore pq = \frac{a}{-a}$$

$$= -1 \quad \text{as req'd.}$$

$$\begin{aligned}
 \text{iii) } PQ &= PS + SQ \\
 &= \sqrt{(2ap)^2 + (ap^2 - a)^2} + \sqrt{(2aq)^2 + (aq^2 - a)^2} \\
 &= \sqrt{4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2} + \sqrt{4a^2q^2 + a^2q^4 - 2a^2q^2 + a^2} \\
 &= \sqrt{a^2(2p^2 + p^4 + 1)} + \sqrt{a^2(2q^2 + q^4 + 1)} \\
 &= \sqrt{a^2(p^2 + 1)^2} + \sqrt{a^2(q^2 + 1)^2} \\
 &= a(p^2 + 1) + a(q^2 + 1) \\
 &= a(p^2 + q^2 + 2) \\
 &= a\left(p^2 + 2 + \frac{1}{p^2}\right) \quad (\text{since } pq = -1) \\
 &= a\left(p + \frac{1}{p}\right)^2 \quad \left(\neq q = -\frac{1}{p} \rightarrow q^2 = \frac{1}{p^2}\right) \\
 &\text{as required}
 \end{aligned}$$

b) i) $y = tx - t^2$

$$\begin{aligned}
 A(x_0, 0) : \quad 0 &= tx - t^2 \\
 \therefore x &= t \\
 \therefore A(t, 0)
 \end{aligned}$$

gradient OT:

$$\begin{aligned}
 m_{OT} &= \frac{t^2}{2t} \\
 &= \frac{t}{2}
 \end{aligned}$$

$$\therefore m_{\perp OT} = -\frac{2}{t}$$

$$\therefore \text{equation AP: } y - 0 = -\frac{2}{t}(x - t)$$

$$\therefore y = -\frac{2}{t}(x - t) \quad \text{as required}$$

ii) equation OT: $y - 0 = \frac{t}{2} (x - 0)$ (1)

$\therefore t = \frac{2y}{x}$ as required

iii) Locus of P $\rightarrow$ simultaneously solve AP & OT:

$y = -\frac{2}{t} (x - t)$ (1)

$t = \frac{2y}{x}$ (2)

concept of simult. solution

Sub (2) into (1):

$y = -\frac{2x}{2y} (x - \frac{2y}{x})$ (1)

$y^2 = -x^2 + 2y$

$x^2 + y^2 - 2y = 0$

$x^2 + y^2 - 2y + 1 = 1$ (1)

$x^2 + (y - 1)^2 = 1$

$\therefore$ locus is circle: centre $(0, 1)$, radius 1 [except the point $(0, 0)$]